Day 2 – Power Supply SystemTheory

1.List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

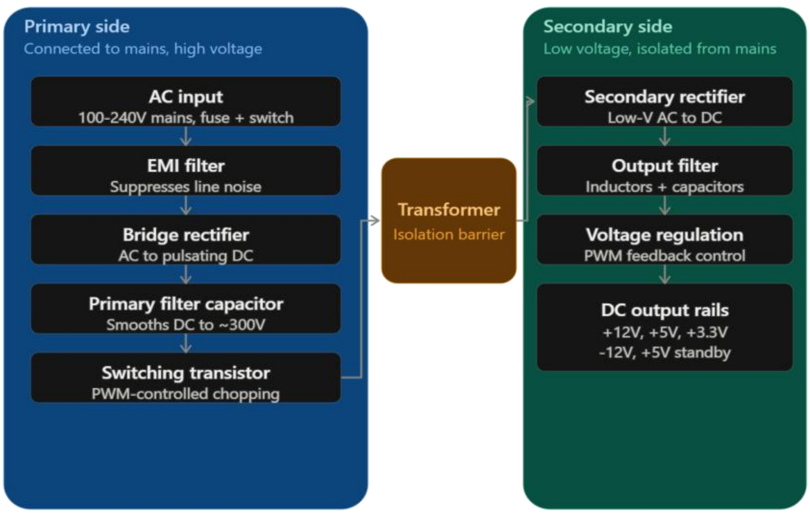
So Here are list of devices use of SMPS

1. Desktop Computer
   We use Smps because it's Converts AC power into the different DC voltage required by computer components
2. Led Tv
   We use Smps because it's provides stable and efficient DC power to the TV's electronic circuits
3. Laptop Charger
   We use Smps because it's convert household AC power into the required low voltage DC power for laptop

Here UPS devices

1. Desktop Computer
   We use UPS because it's provide backup power during power cut so it can work for 10-15 min extra so we can save files
2. Wi-Fi Router
   We use Ups because it's keep the router running during short power failures so the internet connection remain available.

2.Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

ATX 24

Devices it powers - Powers the **motherboard** and provides its main power supply.

Safety Tip-Turn off and unplug the PC before connecting or discc socket**.**

Sata Power Connector

Devices it powers - Powers SATA hard drives, SSDs, optical drives, and some RGB devices.

Safety Tip - Do not force it into the socket; align the connector correctly and disconnect it only when the PC is powered off.

Molex connector
Devices It Powers - Commonly powers older hard drives, optical drives, fans, and other peripherals.
Safety Tip - Hold the plastic connector while removing it—do not pull the wires.

PCIe 6/8-pin connector
Devices It Powers - Provides additional power to graphics cards (GPUs).
Safety Tip - Use the correct PCIe cable and make sure the connector is fully seated before powering on the PC.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Steps to Connect SMPS to a Gaming PC

1 Turn Off the power : Switch off the SMPS and unplug the power cable before starting.

2 Install the SMPS: Place the SMPS in the PC cabinet and secure it with screws.

3 **Connect the motherboard:** Connect the **24-pin ATX cable** from the SMPS to the motherboard's main 24-pin power connector.

4 **Connect the CPU:** Connect the **4+4-pin EPS/CPU power cable** to the CPU power connector near the top of the motherboard.
5 **Connect the graphics card:** Connect the required **6-pin, 8-pin, or 6+2-pin PCIe power cable** from the SMPS to the GPU.
6 **Connect storage devices:** For SATA SSDs or HDDs, connect the **SATA power cable** from the SMPS to the drive.
7 **Check all connections:** Make sure every connector is fully inserted and properly secured.

8 **Organize cables:** Route the SMPS cables through the cabinet's cable-management openings to keep airflow clear.

9 **Connect AC power:** Finally, connect the SMPS's AC power cable to the wall socket and switch on the PC